SIVADINESH PONRAJAN

PhD Researcher in AI Safety and Security

Email | Website | GitHub | LinkedIn | Halmstad, Sweden

RESEARCH PROFILE

PhD researcher working on AI safety, adversarial robustness and mechanistic interpretability. Current research examines explainability and security in multimodal and agentic AI, including adversarial robust vision-language models and unexpected behaviour in autonomous agents.

Research interests: Adversarial robustness, mechanistic interpretability, agent security, weight-space forensics.

EDUCATION

PhD in AI and Cybersecurity *Nov 2025 - Present*
Halmstad University X **Cybercampus Sweden**, Sweden
- **Research project:** *Explainability and Security in Multimodal and Agentic AI*.
- **Supervisors:** Prayag Tiwari (Halmstad University) and Stefan Axelsson (Stockholm University).

MSc in Autonomous Systems *2022 - 2024*
EIT Digital Master School - A double degree master's program
- **Polytech Nice Sophia, Université Côte d'Azur**, **France** - control systems for aerial robotics.
- **Eötvös Loránd University**, **Hungary** - 3D computer vision and vision transformers.
- Secured EIT Digital Master School 50% scholarship for both years and **graduated with honours**.
- **Master's thesis:** *Advancing Soft Real-Time 3D Geometrical Model Construction in Dynamic Environments using Neural Radiance Fields (NeRF): A Comprehensive Exploration and Analysis*.

B.Tech in Information Technology *2016 - 2020*
Sri Sairam Engineering College, Anna University, Chennai, India

RESEARCH EXPERIENCE

PhD Researcher *Nov 2025 - Present*
Halmstad University × Cybercampus Sweden
- Investigating explainability and security in multimodal and agentic AI, with emphasis on adversarially robust vision-language models, reward hacking, task gaming and unexpected agent behaviour.

Research Engineer *Jan - Oct 2025*
Halmstad University × Polisen Sverige
- Worked on *AI-Powered Crime Scene Analysis*, developing soft real-time 3D reconstruction of crime scenes with Neural Radiance Fields.

Master's Thesis Researcher *2023 - 2024*
ELTE Neural Information Processing Research Group, Budapest, Hungary
- Worked on 3D computer vision, vision transformers, drone tracking and soft real-time 3D reconstruction from drone imagery using Neural Radiance Fields.

RESEARCH OUTPUTS

Right Tool, Wrong Arguments: Permission Drift in Quantized Tool-Calling LLMs *2026*
Who Verifies the Agents? Toward Reliable Agent Development - **NeurIPS Workshop - Accepted**
- Examines how quantization can change argument selection and permission-sensitive behaviour in language models that call external tools.

Anonymous Manuscripts Under Review
- Three manuscripts on evaluation gaming in language models, adversarially induced behavioural drift and weight-space based forensics. Titles and authors are withheld during review.

SELECTED RESEARCH PROJECTS

Neural Radiance Fields for Dynamic 3D Reconstruction *2024*
- Developed and analysed soft real-time methods for constructing geometrical models of dynamic environments from drone imagery. Thesis and demonstrations.

LiDAR and Camera Calibration with 3D Sensor Fusion *2024*
- Implemented automatic extrinsic calibration using a spherical target. GitHub.

LLM Reasoning and Retrieval Systems *2024*
- Worked on PDF question answering with retrieval-augmented generation and Llama 2. Developed mathematical reasoning agents using LangChain with chain-of-thought prompting.

Edge AI for Embedded Inference *2023*
- Optimized a convolutional neural network for inference on an STM32 microcontroller. GitHub.

TEACHING AND ACADEMIC SERVICE

Teaching Assistant, Learning Systems
Halmstad University
- Supported course delivery and student learning in Learning Systems.

Ad Hoc Peer Reviewer
- Reviewed private research manuscripts in artificial intelligence and machine learning - venue details remain confidential.

PROFESSIONAL EXPERIENCE

Research Engineer *Jan 2025 - Oct 2025*
Halmstad University × Polisen Sverige, Halmstad, Sweden
- Worked on AI-Powered Crime Scene Analysis, developing soft real-time 3D reconstruction of crime scenes using Neural Radiance Fields (NeRF) from a drone's live video stream.
- Implemented end-to-end 3D computer vision and 3D reconstruction workflows for processing imagery captured in dynamic environments and made post analysis with open-source multimodal AI model (Llava).

Software Engineer *Feb 2021 - Jun 2022*
Prodapt Solutions, Chennai, India
- Worked on the development of telecom applications and python based automations for file-transfer services, including pipeline optimization that improved service speed by 30%.

Research and Engineering Internships *2019 - 2020*
- Worked across cybersecurity, development of OSINT tools, computer vision, Java systems and IoT at Gurugram Cyber Crime Cell, Rheo AI, Zoho and Zion Robotics.

TECHNICAL SKILLS

AI research areas: AI safety, adversarial robustness, mechanistic interpretability, reinforcement learning, supervised fine-tuning, agentic systems, retrieval-augmented generation, prompt engineering, transformers and vision-language models.
Programming: Python, C, C++, Java, shell scripting
Machine learning and Computer Vision: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, OpenCV, LangChain.
Data visualization and monitoring: Matplotlib, Grafana, Tableau.
3D vision, design and simulation: Blender 3D, Fusion 360, Nerfstudio, Visual SLAM, LabVIEW, Scilab, NVIDIA Isaac Sim.
Cloud, web and databases: AWS Bedrock, Amazon SageMaker, Flask, PostgreSQL, MySQL, MongoDB, Firebase.
Robotics and embedded systems: ROS, Docker, STM32, ESP8266, ATmega328, Raspberry Pi, Jetson Nano.

TRAINING AND CERTIFICATIONS

- **BlueDot Impact** - Technical AI Safety, 2026 **(🔗 Link)**
- **NVIDIA** - Exploring Adversarial Machine Learning, 2025 **(🔗 Link)**
- **AWS** - Certified AI Practitioner, 2024 **(🔗 Link)**
- **Udacity** - Flying Car and Autonomous Flight Engineer, 2020 **(🔗 Link)**
- **DeepLearning.AI** - TensorFlow Developer Specialization, 2020 **(🔗 Link)**

LANGUAGES

Tamil: Native proficiency.
English: C1 proficiency, **IELTS** 7.0/9.0
French: A2 proficiency.